The shadow metric, the Riccati algebraicity theorem, and the depth classification

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Remark 0.1 (Relationship to companion papers). The Virasoro spectral R-matrix $R(z) = z^{2h}\exp(-(c/4)/z^2)$ and its connection to the Riccati shadow structure are developed in the companion paper *The Virasoro spectral R-matrix* (`virasoro_r_matrix.tex`).

1 The shadow metric and the Riccati equation

1.1 The shadow obstruction tower

Let $\mathcal{A}$ be a chirally Koszul vertex algebra with modular cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ and universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}))$. The complex carries a degree filtration $F^\bullet$ with $F^r = \bigoplus_{s\geq r} \mathrm{Def}^{(s)}_{\mathrm{cyc}}$, and the *shadow obstruction tower* is the sequence of finite-order truncations

$$\Theta_{\mathcal{A}}^{\leq 2},\ \Theta_{\mathcal{A}}^{\leq 3},\ \Theta_{\mathcal{A}}^{\leq 4},\ \dots$$

where $\Theta_{\mathcal{A}}^{\leq r}$ solves the Maurer–Cartan equation in the quotient $\mathfrak{g}^{\mathrm{amb}}_{\mathcal{A}}/F^{r+1}$. The degree-$r$ projection $\mathrm{Sh}_r(\mathcal{A})$ is the *r-th shadow* of $\mathcal{A}$. On a one-dimensional slice of the deformation complex, Sh_r is determined by a single scalar S_r; the MC equation projected to degree r gives the *all-degree master equation*

$$\nabla_H(\mathrm{Sh}_r) + \mathfrak{o}^{(r)} = 0, \tag{1}$$

where ∇_H is the linearized operator and $\mathfrak{o}^{(r)}$ is the obstruction from lower degrees.

1.2 Primary lines and the shadow metric

Definition 1.1 (Primary line). A *primary line* in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ is a one-dimensional subspace $L = \mathbf{k}\cdot\eta$ spanned by a single primary field η of definite conformal weight h_η. For a chiral algebra with generators $\phi_1, \dots, \phi_N$ of weights $h_1, \dots, h_N$, each generator spans a primary line; the full first-order deformation space is $\bigoplus_i L_i$.

Let L be a primary line with coordinate x, and write $\mathrm{Sh}_r|_L = S_r\, x^r$ for the shadow data restricted to L. Set

$$\kappa := S_2, \qquad \alpha := S_3.$$

The scalar κ is the modular characteristic (curvature) of $\mathcal{A}$ on L; the scalar α is the cubic shadow.

Definition 1.2 (Shadow metric)**.** Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* (shadow quadratic form) on L is the polynomial

$$Q_L(t) \;:=\; a_0^2 + 2\,a_0\,a_1\,t + (a_1^2 + 2\,a_0\,a_2)\,t^2 \;=\; 4\kappa^2 + 12\kappa\alpha\,t + (9\alpha^2 + 16\kappa S_4)\,t^2 \;\in\; k(c)[t]. \quad (2)$$

The *critical discriminant* of L is

$$\Delta := 8\kappa S_4 = a_0\,a_2.$$

Remark 1.3. The shadow metric Q_L is a quadratic polynomial in t, determined by three OPE invariants: κ, α, S_4. The term "metric" refers to Q_L's role as the quadratic form governing the tower's geometry (branch points, monodromy, convergence radius), not to a Riemannian metric.

1.3 The recursion on a primary line

On a primary line L, the all-degree master equation (1) reduces to a recursion among the scalars S_r.

Proposition 1.4 (Single-line recursion)**.** *On a primary line L with propagator $P := 1/\kappa$, the MC equation at degree $r \geq 5$ is*

$$S_r \;=\; -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3\leq j\leq k}} c_{jk}\; j\,k\,S_j\,S_k, \qquad c_{jk} = \begin{cases} 1 & \text{if } j<k, \\ 1/2 & \text{if } j=k. \end{cases} \quad (3)$$

This recursion is well-defined: since $j \geq 3$ and $j + k = r + 2$, every term on the right has $k = r + 2 - j \leq r - 1$, so S_r is determined by $\{S_2, S_3, S_4, \ldots, S_{r-1}\}$. The degree-4 shadow S_4 is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside $\kappa = S_2$ and $\alpha = S_3$.

Proof. The Lie bracket on the cyclic deformation complex, restricted to L, pairs $\mathrm{Sh}_j = S_j x^j$ and $\mathrm{Sh}_k = S_k x^k$ to produce a term of degree $j + k - 2$ with coefficient $j k\, P\, S_j S_k$. The eigenvalue identity $\nabla_H(x^r) = 2r\,x^r$ separates the MC equation at degree r into the linear term $2r\,S_r$ on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor c_{jk} accounts for the factor $\frac{1}{2}$ in $\frac{1}{2}[\Theta,\Theta]$ when $j = k$. □

1.4 The weighted shadow generating function

Definition 1.5 (Weighted shadow generating function)**.** The *weighted shadow generating function* on L is

$$H(t) \;:=\; \sum_{r\geq 2} r\,S_r\,t^r.$$

Setting $a_n := (n + 2)\,S_{n+2}$ for $n \geq 0$, this becomes $H(t) = \sum_{n\geq 0} a_n\,t^{n+2}$, with initial values $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$.

The weighted generating function H packages the recursion (3) as a convolution identity. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\,t^n$. The MC recursion forces every Cauchy coefficient of F^2 beyond degree 2 to vanish, so F^2 is a polynomial of degree 2.

2 Algebraicity and the depth classification

2.1 The Riccati algebraicity theorem

Theorem 2.1 (Riccati algebraicity). *Let L be a primary line with shadow data S_r, shadow metric Q_L (Definition 1.2), and weighted shadow generating function H (Definition 1.5). Then the all-degree master equation on L is equivalent to the algebraic relation*

$$H(t)^2 \;=\; t^4\, Q_L(t). \tag{4}$$

The shadow generating function is therefore algebraic of degree 2 over $k(\kappa, \alpha, S_4)(t)$:

$$H(t) \;=\; t^2\sqrt{Q_L(t)}.$$

Proof. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\, t^n$ where $a_n = (n+2)\, S_{n+2}$. The claim (4) is equivalent to $F(t)^2 = Q_L(t)$. We verify this by computing the Cauchy product $F^2 = \sum_{m\geq 0}\big(\sum_{i+j=m} a_i a_j\big)\, t^m$ and showing it agrees with Q_L at all orders.

Orders 0, 1, 2 *(initial data).* At $m = 0$: $a_0^2 = (2\kappa)^2 = 4\kappa^2$. At $m = 1$: $2\, a_0\, a_1 = 2\cdot 2\kappa \cdot 3\alpha = 12\kappa\alpha$. At $m = 2$: $a_1^2 + 2\, a_0\, a_2 = 9\alpha^2 + 2\cdot 2\kappa\cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$. These are exactly the coefficients of $Q_L(t)$ in (2).

Orders $m \geq 3$ (the MC recursion forces vanishing). Since Q_L has degree 2 in t, the identity $F^2 = Q_L$ requires

$$\sum_{i+j=m} a_i\, a_j \;=\; 0 \qquad \text{for all } m \geq 3. \tag{5}$$

We show that this is exactly the MC recursion (3) at degree $r = m + 2$.

The sum $\sum_{i+j=m} a_i a_j$ with $a_i = (i+2)\, S_{i+2}$ and $a_j = (j+2)\, S_{j+2}$ equals $\sum_{p+q=m+4,\; p,q\geq 2} p\, q\, S_p\, S_q$, where $p = i+2$, $q = j+2$. Setting $r = m+2$, this becomes $\sum_{p+q=r+2,\; p,q\geq 2} p\, q\, S_p\, S_q$.

Isolating the terms with $p = 2$ or $q = 2$ (which contribute 2κ from $S_2 = \kappa$):

$$\begin{aligned}\sum_{p+q=r+2,\; p,q\geq 2} pq\, S_p S_q &= 2\cdot 2r\cdot \kappa\, S_r + \sum_{\substack{p+q=r+2\\ p,q\geq 3}} pq\, S_p S_q\\ &= 4r\kappa\, S_r + 2\sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_j S_k,\end{aligned}$$

where the factor 2 in front of the restricted sum accounts for $S_pS_q + S_qS_p$ when $p \neq q$, and c_{jk} corrects for the diagonal. Substituting the recursion (3):

$$2\sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_j S_k \;=\; 2\cdot(-2r\kappa\, S_r) \;=\; -4r\kappa\, S_r.$$

Hence $\sum_{p+q=r+2} pq\, S_pS_q = 4r\kappa S_r - 4r\kappa S_r = 0$, which is (5).

Conversely, the vanishing (5) at order $m \geq 3$ implies the recursion (3) at $r = m+2$ by solving for S_r (using $\kappa \neq 0$).

The equivalence of (4) with the full MC recursion is therefore established at all degrees. Taking positive square roots (the branch with $F(0) = a_0 = 2\kappa > 0$) gives $H(t) = t^2 F(t) = t^2\sqrt{Q_L(t)}$. □

Remark 2.2 (Riccati ODE reformulation). Set $U(t) := \sum_{r\geq 3} S_r\, t^r$ (the nonlinear part, removing Sh_2). The algebraic relation (4) is equivalent to the Riccati-type first-order ODE

$$2t\, U'(t) + \frac{P}{2}\big(U'(t)\big)^2 = R_3\, t^3 + R_4\, t^4,$$

where $R_3 = 6\alpha$ and $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$ encode the cubic and quartic inputs. The left-hand side is quadratic in U': this is the precise sense in which the shadow obstruction tower is algebraic of degree 2.

2.2 Gaussian decomposition

Corollary 2.3 (Gaussian decomposition)**.** *The shadow metric decomposes as*

$$Q_L(t) \;=\; (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2,$$

where $\Delta = 8\kappa S_4$ is the critical discriminant. The first term $(2\kappa + 3\alpha t)^2$ is the Gaussian envelope*: the shadow metric of the truncated tower with $S_r = 0$ for $r \geq 4$. The second term $2\Delta\, t^2$ is the* interaction correction*, measuring the quartic OPE contribution not captured by the cubic data alone.*

Proof. $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha\, t + 9\alpha^2\, t^2$. Adding $2\Delta\, t^2 = 16\kappa S_4\, t^2$ recovers the coefficient $9\alpha^2 + 16\kappa S_4$ of t^2 in $Q_L(t)$. □

2.3 Intrinsic quartic principle

Corollary 2.4 (Intrinsic quartic principle)**.** *On a primary line L, the shadow obstruction tower has intrinsic OPE data at degrees 2, 3, 4 only. All shadow coefficients S_r for $r \geq 5$ are determined by (κ, α, S_4) alone, via the closed form*

$$S_r \;=\; \frac{1}{r}\,[t^{r-2}]\,\sqrt{Q_L(t)} \qquad (r \geq 2).$$

Proof. $Q_L(t) = 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$ depends on exactly (κ, α, S_4). Since $H(t) = t^2\sqrt{Q_L(t)} = \sum_{r\geq 2} rS_r\, t^r$ by Theorem 2.1, comparing coefficients gives $rS_r = [t^{r-2}]\sqrt{Q_L}$. □

2.4 The shadow connection

Proposition 2.5 (Shadow connection)**.** *The shadow metric defines a logarithmic connection on $\mathcal{O}_L$ over $L \setminus \{Q_L = 0\}$:*

$$\nabla^{\mathrm{sh}} \;=\; d - \omega, \qquad \omega \;:=\; \frac{Q_L'(t)}{2\, Q_L(t)}\, dt \;=\; d\big(\log \sqrt{Q_L}\big).$$

(i) *The connection is flat:* $(\nabla^{\mathrm{sh}})^2 = 0$.

(ii) *At each simple zero t_0 of Q_L, the residue of ω is $\frac{1}{2}$.*

(iii) *The monodromy is* $\exp(2\pi i \cdot \frac{1}{2}) = -1$*: the standard monodromy of a square root around a simple branch point.*

(iv) *The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)}/(2\kappa)$, and the shadow generating function is $H(t) = 2\kappa\, t^2\, \Phi(t)$: the parallel transport of the curvature along the degree parameter, weighted by t^2.*

Proof. Flatness is automatic: ω is a closed 1-form on a 1-dimensional base. At a simple zero t_0 of Q_L: $Q_L(t) \approx Q_L'(t_0)(t - t_0)$, so $\omega \approx \frac{1}{2(t-t_0)}\, dt$, giving residue $\frac{1}{2}$. The monodromy exponentiation is immediate. For (iv): $\nabla^{\mathrm{sh}}\Phi = 0$ since $d\Phi = \omega\,\Phi$ by construction, and the evaluation at $t = 0$ gives $\Phi(0) = \sqrt{Q_L(0)}/(2\kappa) = 2\kappa/(2\kappa) = 1$. □

Remark 2.6. The shadow generating function $H(t) = 2\kappa\, t^2\, \Phi(t)$ is *not* a flat section of ∇^{sh}: the factor t^2 contributes $\nabla^{\mathrm{sh}}(H) \neq 0$. The flat section is $\Phi(t) = \sqrt{Q_L(t)}/(2\kappa)$; the shadow obstruction tower is the Taylor expansion of $t^2 \cdot \Phi(t)$, reflecting the degree offset (the tower starts at degree 2, not degree 0).

2.5 The single-line dichotomy

Theorem 2.7 (Single-line dichotomy). *Let L, S_r, α, Δ, Q_L be as above, and define the shadow depth $r_{\max}|_L := \max\{r : S_r \neq 0\}$ (with $r_{\max} = \infty$ if $S_r \neq 0$ for infinitely many r). Then $r_{\max}|_L \in \{2, 3, \infty\}$, classified by the shadow metric:*

- (i) ***Class* G** *($\Delta = 0$, $\alpha = 0$). $Q_L = (2\kappa)^2$ is constant. $H(t) = 2\kappa\, t^2$. Only S_2 survives: $r_{\max} = 2$.*
- (ii) ***Class* L** *($\Delta = 0$, $\alpha \neq 0$). $Q_L = (2\kappa + 3\alpha\, t)^2$ is a perfect square. $H(t) = t^2(2\kappa + 3\alpha\, t)$. S_2 and S_3 survive: $r_{\max} = 3$.*
- (iii) ***Class* M** *($\Delta \neq 0$). Q_L is irreducible over $k(c)$; $\sqrt{Q_L}$ is irrational; $S_r \neq 0$ for all $r \geq 4$ generically: $r_{\max} = \infty$.*

For $r \geq 4$,

$$S_r = \Delta \cdot R_r, \qquad R_r \in \mathbb{Q}(\alpha, \kappa, S_4).$$

Proof. The classification follows from Theorem 2.1 and the Gaussian decomposition (Corollary 2.3).

Case $\Delta = 0$. Then $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square, so $\sqrt{Q_L} = 2\kappa + 3\alpha t$ is polynomial and $H(t) = t^2(2\kappa + 3\alpha t)$ terminates. If $\alpha = 0$: $H(t) = 2\kappa t^2$, giving $r_{\max} = 2$. If $\alpha \neq 0$: $H(t) = 2\kappa t^2 + 3\alpha t^3$, giving $r_{\max} = 3$.

Case $\Delta \neq 0$. The discriminant of Q_L as a polynomial in t is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so Q_L is irreducible over $k(c)$. The Taylor expansion of $\sqrt{Q_L}$ around $t = 0$ has nonzero coefficients at all orders generically, hence $r_{\max} = \infty$.

Universal factorization. The binomial expansion of $\sqrt{Q_L}$ around the Gaussian envelope $(2\kappa + 3\alpha t)$ is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t)\sqrt{1 + \frac{2\Delta\, t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding $\sqrt{1+u}$ as $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \cdots$ with $u = 2\Delta t^2/(2\kappa + 3\alpha t)^2$, every coefficient of t^n for $n \geq 2$ in $\sqrt{Q_L} - (2\kappa + 3\alpha t)$ carries a factor of Δ. Since $rS_r = [t^{r-2}]\sqrt{Q_L}$ for $r \geq 2$, and the Gaussian part $2\kappa + 3\alpha t$ contributes only at $r \leq 3$, the factorization $S_r = \Delta \cdot R_r$ for $r \geq 4$ follows. □

Remark 2.8 (The contact class). The partition **G**/**L**/**C**/**M** of chirally Koszul algebras includes a fourth class **C** (contact, $r_{\max} = 4$), realized by the $\beta\gamma$ system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}$ where $\kappa = 0$, and the shadow metric requires $\kappa \neq 0$. On the charged stratum, the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at $r = 4$. The algebraic shadow depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ exhausts the four classes: **G** ($d_{\mathrm{alg}} = 0$), **L** ($d_{\mathrm{alg}} = 1$), **C** ($d_{\mathrm{alg}} = 2$), **M** ($d_{\mathrm{alg}} = \infty$).

2.6 The shadow growth rate

For class **M** algebras, the shadow obstruction tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients S_r.

Definition 2.9 (Shadow growth rate). Let L be a primary line of class **M** ($\Delta \neq 0$), and write $Q_L(t) = q_0 + q_1 t + q_2 t^2$ for the shadow metric. The *shadow growth rate* of L is

$$\rho_L \ := \ \sqrt{\frac{q_2}{q_0}} \ = \ \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

By Vieta's theorem, the zeros $t_0, \bar{t}_0$ of Q_L satisfy $|t_0|^2 = q_0/q_2$, so $\rho_L = 1/|t_0|$: the reciprocal of the branch-point modulus. For classes **G** and **L**, set $\rho := 0$ by convention.

Theorem 2.10 (Shadow asymptotics). *Let L be a primary line of class* **M** *with shadow growth rate ρ_L.*

(i) Asymptotic formula.

$$S_r \ \sim \ A(\mathcal{A})\, \rho_L^r\, r^{-5/2} \cos(r\theta + \varphi) \qquad (r \to \infty),$$

where $\theta := -\arg(t_0)$ is the oscillation phase, $A(\mathcal{A}) > 0$, and φ is computable from the branch-point residue of $\sqrt{Q_L}$. The exponent $-5/2$ arises from $S_r = [t^{r-2}]\sqrt{Q_L}/r$ combined with the $n^{-3/2}$ transfer law for square-root singularities.

(ii) Unit circle criterion. *$\rho_L = 1$ if and only if $q_0 = q_2$, i.e.,*

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta.$$

Equivalently, the branch points of Q_L lie on the unit circle $|t| = 1$.

(iii) Convergence partition. *$\rho_L < 1$: the shadow obstruction tower converges absolutely. $\rho_L > 1$: the shadow obstruction tower diverges, but the algebraic function $\sqrt{Q_L(t)}$ extends analytically past the convergence radius. $\rho_L = 1$: marginal case (the critical surface).*

Proof. The function $\sqrt{Q_L(t)}$ has algebraic branch-cut singularities of order $\frac{1}{2}$ at each simple zero t_0 of Q_L. Since Q_L is quadratic with nonzero discriminant ($\Delta \neq 0$ implies $\mathrm{disc}(Q_L) = -32\kappa^2\Delta \neq 0$), the two zeros $t_0, \bar{t}_0$ are distinct with $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$.

The transfer theorem for algebraic singularities [?FlajoletSedgewick, Theorem VI.1] gives $[t^n]\sqrt{Q_L} \ \sim \ C_0\, t_0^{-n}\, n^{-3/2}$ for each dominant singularity t_0 at distance $R = 1/\rho_L$ from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from $2\,\mathrm{Re}(C_0\, t_0^{-(r-2)})$. Since $S_r = [t^{r-2}]\sqrt{Q_L}/r$ (Corollary 2.4), the extra factor $1/r$ gives the stated $r^{-5/2}$ exponent.

For (ii): $\rho_L = 1$ iff $q_2/q_0 = 1$. Writing $q_0 = 4\kappa^2$ and $q_2 = 9\alpha^2 + 2\Delta$, the condition $q_0 = q_2$ factors as $4\kappa^2 - 9\alpha^2 = 2\Delta$. Part (iii) is immediate from the definition of the convergence radius $R = 1/\rho_L$. ☐

2.7 The depth decomposition

Theorem 2.11 (Depth decomposition). *The shadow depth of a chirally Koszul algebra $\mathcal{A}$ decomposes as*

$$d(\mathcal{A}) \ = \ 1 + d_{\mathrm{arith}}(\mathcal{A}) + d_{\mathrm{alg}}(\mathcal{A}),$$

where:

(i) d_{arith} *counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function* $\widehat{Z}^c_{\mathcal{A}}$ *on* $\mathcal{M}_{1,1}$*;*

(ii) $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ *is the* algebraic depth*, determined by the single-line dichotomy* (*Theorem* 2.7) *and stratum separation* (*Remark* 2.8)*:*

$$d_{\mathrm{alg}} = \begin{cases} 0 & \textit{class } \mathbf{G}, \\ 1 & \textit{class } \mathbf{L}, \\ 2 & \textit{class } \mathbf{C}, \\ \infty & \textit{class } \mathbf{M}; \end{cases}$$

(iii) *depths* $d \geq 5$ *are purely arithmetic: they arise from* d_{arith}*, not from the* A_∞ *structure.*

Proof. The algebraic depth d_{alg} is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 2.7) gives $d_{\mathrm{alg}} \in \{0, 1, \infty\}$ on each primary line, and stratum separation (Remark 2.8) provides $d_{\mathrm{alg}} = 2$ for the contact class. Since $d_{\mathrm{alg}} \leq 2$ for finite cases ($\mathbf{G}/\mathbf{L}/\mathbf{C}$) and the base term is 1 (the curvature κ at degree 2), the maximum finite depth without arithmetic contributions is $1 + 0 + 2 = 3$; hence all depths $d \geq 5$ require $d_{\mathrm{arith}} \geq 2$.

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing $\widehat{Z}^c_{\mathcal{A}}$ via the Roelcke–Selberg spectral decomposition on $\mathcal{M}_{1,1}$. Each holomorphic eigenform of weight k contributes a functional-equation centre at $\mathrm{Re}(s) = k/2$; Maass cusp-form projections contribute no new constituent centres. □

Example 2.12 (Standard families)**.** The shadow metric data for the standard families:

Family	κ	α	S_4	Δ	Class	ρ
Heisenberg $\mathcal{H}_k$	k	0	0	0	$\mathbf{G}$	0
Affine $\widehat{\mathfrak{g}}_k$	$\frac{\dim\mathfrak{g}\,(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	$\mathbf{L}$	0
$\beta\gamma$ (weight λ)	$6\lambda^2{-}6\lambda{+}1$	0^*	0^*	0^*	$\mathbf{C}$	0
Virasoro Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	$\mathbf{M}$	$\frac{1}{c}\sqrt{\frac{180c+872}{5c+22}}$

* The table reports the weight-changing primary line, where $\kappa = S_3 = S_4 = 0$. The quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on this charged stratum; the shadow metric requires $\kappa \neq 0$ and does not apply. The class $\mathbf{C}$ designation ($r_{\max} = 4$) comes from rank-one rigidity, not the single-line dichotomy. The T-line (stress tensor) carries the Virasoro shadow data $(\kappa, \alpha, S_4) = (c/2, 2, 10/(c(5c{+}22)))$ with $c = 2(6\lambda^2{-}6\lambda{+}1)$ and is class $\mathbf{M}$.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\mathrm{Vir}}(t) = c^2 + 12c\,t + \frac{180c + 872}{5c + 22}\,t^2,$$

with Gaussian decomposition $Q_{\mathrm{Vir}} = (c + 6t)^2 + \frac{80}{5c+22}\,t^2$. The critical central charge c^* where $\rho = 1$ is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately $c^* \approx 6.12$. For $c > c^*$ the shadow tower converges; for $c < c^*$ it diverges. At the Koszul self-dual point $c = 13$: $\rho(\mathrm{Vir}_{13}) \approx 0.467$.

Remark 2.13 (Large-c asymptotic of S_4). The Virasoro quartic shadow

$$S_4(\mathrm{Vir}_c) \;=\; \frac{10}{c\,(5c+22)}$$

has large-c asymptotic

$$S_4(\mathrm{Vir}_c) \;=\; \frac{10}{5c^2}\Big(1 - \tfrac{22}{5c} + O(c^{-2})\Big) \;=\; \frac{2}{c^2} + O(c^{-3}),$$

not $2/(5c^2)$; the factor of 5 in the denominator of $10/[c(5c+22)]$ is absorbed when expanding $1/(5c+22)$ at large c. Pairing this with $\kappa = c/2$ gives the critical discriminant $\Delta = 8\kappa S_4 = 40/(5c+22) \longrightarrow 8/c$ as $c \to \infty$, confirming class **M** mixing persists to leading order.

The identity $S_2(\mathrm{Vir}_c) = \kappa = c/2$ is convention-independent. The occasional error $S_2 = c/12$ conflates the shadow invariant with the Vol II lambda-bracket divided-power coefficient $c/12 = (c/2)/3!$, which is a different normalisation; the Vol I–II bridge is $S_2^{\mathrm{Vol\,I}} = 6\,S_2^{\lambda}$.

The Riccati seed on Vir_c is $(r_0, S_{r_0}) = (3, 2)$: $S_3(\mathrm{Vir}_c) = 2$ is the c-independent cubic shadow (Lorgat, *The Virasoro spectral R-matrix*), and the shadow-exponential base $C_A = 6$ on the non-logarithmic class **M** locus factorises as $6 = r_0 \cdot S_{r_0} = 3 \cdot 2$.

Remark 2.14 (Spectral curve). The algebraic relation $H^2 = t^4 Q_L(t)$ defines, for each primary line L, a spectral curve $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}^2_{t,H}$. Since Q_L is quadratic in t, Σ_L has arithmetic genus 0; the projection $(t, H) \mapsto t$ is a double cover ramified at the zeros of Q_L. The degeneration loci in parameter space are controlled by the critical discriminant: Σ_L acquires a node precisely when $\Delta = 0$. The shadow growth rate ρ_L is the reciprocal of the distance from $t = 0$ to the nearest ramification point.